

APPENDIX B

Classifying Spaces and Characteristic Classes

The point of this appendix is to briefly summarize the basic facts from the theory of classifying spaces and characteristic classes for principal G -bundles. Details can be found in Milnor-Stasheff [1] and Husemoller [1].

For simplicity we shall assume throughout this appendix that G is a Lie group. Furthermore all spaces will be assumed to have base points, and all maps will be base-point-preserving. All spaces will also be assumed to have the homotopy type of a countable CW-complex.

DEFINITION B.1. A **classifying space** for the group G is a connected topological space BG , together with a principal G -bundle $EG \rightarrow BG$, such that the following is true. For any compact Hausdorff space X , there is a one-to-one correspondence between the equivalence classes of principal G -bundles on X and the homotopy classes of maps from X to BG , given by associating to each map $f: X \rightarrow BG$, the induced bundle f^*EG over X (see the end of App. A).

Thus the induced bundle construction gives a natural bijection

$$\text{Prin}_G(X) \cong [X, BG]. \quad (\text{B.1})$$

Note that as a special case we have

$$\text{Prin}_G(S^n) \cong \pi_n(BG). \quad (\text{B.2})$$

Given two different classifying spaces for G , say B_0G and B_1G , there must exist mappings $f_0: B_0G \rightarrow B_1G$ and $f_1: B_1G \rightarrow B_0G$ such that $f_i^*E_{i+1}G \cong E_iG$ for $i \in \mathbb{Z}_2$. This implies that $(f_{i+1} \circ f_i)^*E_iG \cong E_iG$, and therefore $f_{i+1} \circ f_i$ is homotopic to the identity on B_iG for $i \in \mathbb{Z}_2$. Thus B_0G and B_1G are homotopy equivalent, i.e., BG is well defined up to homotopy type.

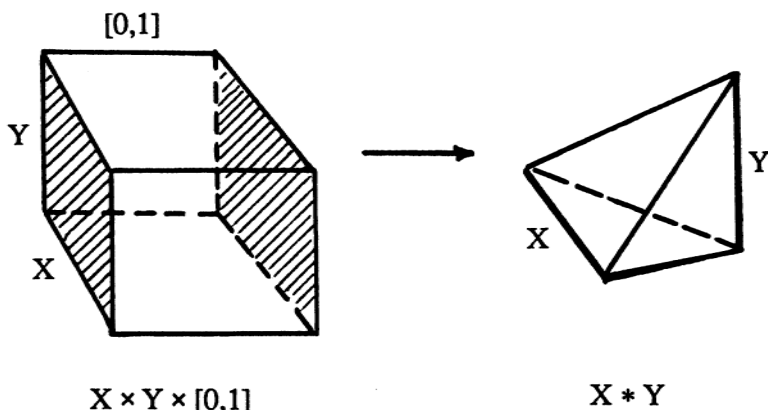
The bundle $EG \rightarrow BG$ is called the **universal principal G -bundle**. This bundle has the following homotopy characterization (see Steenrod [1] for example).

Theorem B.2. *Let $E \rightarrow B$ be a principal G -bundle with the property that the total space of E is contractible. Then (B, E) is a classifying space for G .*

The theory has definite interest due to the following:

Theorem B.3. *For any Lie group there exists a classifying space.*

The general construction of classifying spaces for principal bundles is due to Milnor [1,2] and proceeds as follows. Given a group G , we consider the n -fold join $G * \cdots * G$. (The **join** of two spaces X and Y is obtained from $X \times Y \times [0,1]$ by collapsing $\{x\} \times Y \times \{0\}$ to a point for each x and also collapsing $X \times \{y\} \times \{1\}$ to a point for each y .)



$$X \times Y \times [0,1] \longrightarrow X * Y$$

The space $G * \cdots * G$ is $(n-1)$ -connected. It also has a free G -action obtained by multiplying simultaneously on the right in each of the factors. We pass to the limit as $n \rightarrow \infty$, and define $EG \equiv G * G * G * \cdots$. This is infinitely-connected and has a free right G -action. We set $BG = EG/G$ with $\pi: EG \rightarrow BG$ the quotient map. From Theorem B.2, this bundle is classifying.

Of course this construction of Milnor can (and has) been carried out in enormous generality.

It should be noted that since EG is contractible, the long exact sequence of homotopy groups for a fibration implies that

$$\pi_{n-1}(G) \cong \pi_n(BG) \quad (\text{B.3})$$

for all $n \geq 1$. Thus, the homotopy groups of BG present nothing new. However, the homology of the space is quite interesting.

We begin the discussion with the following. Consider the singular cohomology $H^*(BG; \Lambda)$ with coefficients in a ring Λ .

DEFINITION B.4. Each non-zero class in $H^*(BG; \Lambda)$ is a **universal characteristic class** for principal G -bundles.

Fix a class $c \in H^*(BG; \Lambda)$. Then for any principal G -bundle $P \rightarrow X$ there is a map $f_P: X \rightarrow BG$ so that $P \cong f_P^* EG$. We define the **c-characteristic**

class of P to be the class

$$c(P) = f_P^*(c) \in H^k(X; \Lambda). \quad (\text{B.4})$$

Since f_P is well defined up to homotopy, the class $c(P)$ is uniquely defined.

Observe that any such characteristic class transforms naturally in the following sense. Given a principal G -bundle $P \rightarrow X$ and a continuous map $F: Y \rightarrow X$, the c -characteristic classes satisfy:

$$c(F^*P) = F^*c(P) \quad (\text{B.5})$$

To see this we simply note that $f_{F^*P} = f_P \circ F$, and so $c(F^*P) = f_{F^*P}^*(c) = F^*f_P^*(c) = F^*c(P)$.

Given a continuous homomorphism $\varphi: H \rightarrow G$ of Lie groups, there is a corresponding continuous map

$$B\varphi: BH \longrightarrow BG \quad (\text{B.6})$$

which classifies the principal G -bundle over BH associated to the representation φ . That is,

$$(B\varphi)^*(EG) = EH \times_{\varphi} G$$

where $EH \times_{\varphi} G$ denotes the quotient of $EH \times G$ by the action of H given by setting $\Phi_h(p, g) \equiv (ph^{-1}, hg)$ for $p \in EH$, $g \in G$ and $h \in H$. This association also transforms naturally. That is, for any composite homomorphism $H \xrightarrow{\psi} K \xrightarrow{\varphi} G$, we have

$$B(\psi \circ \varphi) = B(\psi) \circ B(\varphi).$$

If the homomorphism $\varphi: H \rightarrow G$ is a homotopy equivalence, then so is the mapping $B\varphi: BH \rightarrow BG$. This is easily seen by applying the 5-lemma to the induced map on the long exact homotopy sequences of the fibrations

$$H \longrightarrow EH \longrightarrow BH \quad \text{and} \quad G \longrightarrow EG \longrightarrow BG.$$

(Recall that BH and BG are assumed to be homotopy equivalent to countable CW-complexes.) This gives the following result:

Proposition B.5. *Let G be a connected Lie group and $K \subseteq G$ a maximal compact subgroup. Then $BG \cong BK$, that is, the classifying spaces of G and K are homotopy equivalent.*

Proof. It follows from the Iwasawa decomposition of G (cf. Helgason [1]) that the inclusion $K \hookrightarrow G$ is a homotopy equivalence. ■

It follows, for example, that $BO_n \cong BGL_n(\mathbb{R})$, $BSO_n \cong BGL_n^+(\mathbb{R})$, $BU_n \cong BGL_n(\mathbb{C})$, $BSU_n \cong BSL_n(\mathbb{C})$, $BSp_n \cong BSp_n(\mathbb{C})$, etc.

Note that, in general, each mapping of the type $B\varphi: BH \rightarrow BG$ induces a transformation $(B\varphi)^*: H^*(BG; \Lambda) \rightarrow H^*(BH; \Lambda)$ of universal characteristic classes. Any non-zero element which goes to zero under this transformation is a universal obstruction to the reduction of the structure group from G to H . That is, if c is such a class and if $P \rightarrow X$ is a principal G -bundle with $c(P) \neq 0$, then P cannot be written as an associated bundle $P = P' \times_{\varphi} G$ (as above) for any principal H -bundle $P' \rightarrow X$.

The cases of basic interest here are those of the compact classical groups. For these cases there are useful alternative constructions of the classifying spaces.

For $K = \mathbb{R}, \mathbb{C}$ or $\mathbb{H}$, let $G^K(n, N)$ denote the Grassmann manifold of all n -dimensional K -linear subspaces of K^N . There are natural inclusions $G^K(n, N) \subset G^K(n, N+N')$ induced by the "first-coordinate" inclusions $K^N \subset K^{N+N'}$. Taking the direct limit, we get

$$G^K(n, \infty) \equiv \lim_{N \rightarrow \infty} G^K(n, N),$$

the Grassmannian of K -planes in K^∞ .

There is a **canonical K -vector bundle** $E_n \rightarrow G^K(n, \infty)$ whose fibre at a plane $P \in G^K(n, \infty)$ consists of all the vectors in that plane. We shall show that

$$\begin{aligned} BO_n &\cong G^{\mathbb{R}}(n, \infty) \\ BU_n &\cong G^{\mathbb{C}}(n, \infty) \\ BSp_n &\cong G^{\mathbb{H}}(n, \infty). \end{aligned} \tag{B.7}$$

The principal fibrations in each case are obtained by taking the appropriate bundle of bases (orthonormal, unitary, or symplectic) in the canonical bundle E_n . Thereby, E_n becomes the **universal (real, complex or quaternionic) n -plane bundle** over the classifying space (BO_n , BU_n , or BSp_n respectively).

We begin our proof of these claims with the real case. For this we note that for any N , we have $G^{\mathbb{R}}(n, N) \cong O_N/(O_{N-n} \times O_n)$. The bundle of orthonormal frames for the canonical n -plane bundle $E_n \rightarrow G^{\mathbb{R}}(n, N)$ is just the Stiefel manifold O_N/O_{N-n} . Hence, the bundle of orthonormal frames for $E_n \rightarrow G^{\mathbb{R}}(n, \infty)$ is just the direct limit over N of the principal O_n -bundles

$$O_N/O_{N-n} \longrightarrow O_N/(O_{N-n} \times O_n)$$

It is an elementary exercise, using the fibrations $O_k \rightarrow O_{k+1} \rightarrow S^k$, to see that O_N/O_{N-n} is $(N-n-2)$ -connected. Hence the infinite Stiefel manifold $\varinjlim_N O_N/O_{N-n}$ is contractible, and the limiting fibration must represent the universal bundle $EO_n \rightarrow BO_n$ by Theorem B.2.

The arguments for BU_n and BSp_n are entirely analogous. Here one uses the facts that $G^c(n, N) \cong U_N/(U_{N-n} \times U_n)$ and $G^h(n, N) \cong Sp_N/(Sp_{N-n} \times Sp_n)$. We omit the details.

The reader has undoubtedly noticed that BO_n can be considered as the classifying space for real n -dimensional vector bundles. That is, for a compact Hausdorff space X , the equivalence classes of such bundles correspond one-to-one with elements of $[X, BO_n]$ by associating to $f: X \rightarrow BO_n$ the vector bundle f^*E_n . This follows easily from the discussion above by passing to the bundles of orthonormal bases. The corresponding remarks hold for BU_n and BSP_n .

It should be noted that the fact that $G^K(n, \infty)$ is a classifying space for n -dimensional vector bundles (over K) can be proved by a direct geometric construction (see Milnor-Stasheff [1]).

We now examine the cohomology of the basic classifying spaces. These spaces are infinite-dimensional and have non-zero classes in arbitrarily high dimensions. Nevertheless, as **rings under the cup-product multiplication**, they are quite understandable. Basic references for all the following results are Milnor-Stasheff [1], Steenrod-Epstein [1], and Borel [2]. We begin with $\mathbb{Z}_2$ -cohomology.

Theorem B.7. *The cohomology ring $H^*(BO_n; \mathbb{Z}_2)$ is a $\mathbb{Z}_2$ -polynomial ring*

$$\mathbb{Z}_2[w_1, w_2, \dots, w_n]$$

on canonical generators $w_k \in H^k(BO_n; \mathbb{Z}_2)$ for $k = 1, \dots, n$.

Note. The classes $w_1, \dots, w_n$ are characterized inductively by the following fact. The kernel of the homomorphism $H^*(BO_n; \mathbb{Z}_2) \xrightarrow{i^*} H^*(BO_{n-1}; \mathbb{Z}_2)$ induced by the standard inclusion $O_{n-1} \xrightarrow{i} O_n$, is the principal ideal $\langle w_n \rangle$. Analogous statements characterize the canonical generators in the subsequent theorems.

The class w_k is called the **universal k th Stiefel-Whitney class**. Thus to any n -dimensional real vector bundle $E \rightarrow X$ classified by a map $f_E: X \rightarrow BO_n$, we have the associated k th Stiefel-Whitney class of E , $w_k(E) \equiv f_E^*(w_k)$. An important concept is that of the **total Stiefel-Whitney class** $w \equiv 1 + w_1 + \dots + w_n$. It satisfies the Whitney product formula for the sum of two bundles

$$w(E \oplus E') = w(E) \cup w(E') \quad (\text{B.8})$$

which translates in longhand to the equations

$$w_k(E \oplus E') = \sum_{i=0}^k w_i(E) \cup w_{k-i}(E')$$

for $k = 1, 2, \dots$. This can be considered as a statement about the induced map on cohomology induced by $B(O_n \times O_n) \rightarrow BO_{n+n}$ under the usual "diagonal block" inclusion $O_n \times O_n \subset O_{n+n}$.

Given a smooth manifold X , we define the k th Stiefel-Whitney class of X to be $w_k(TX)$. It is an important fact that the *Stiefel-Whitney classes of a compact smooth manifold are invariants of the homotopy type of the manifold*. That is, given a homotopy equivalence $f: Y \rightarrow X$ between compact smooth manifolds, we have that $f^*w_k(X) = w_k(Y)$ (see Milnor-Stasheff [1], p. 131.) This will not be true of many other characteristic classes of manifolds.

Consider now the space BSO_n . This can be realized as above as the Grassmannian of oriented n -planes in $\mathbb{R}^\infty$. There are natural maps $BSO_n \rightarrow BO_n \rightarrow B\mathbb{Z}_2 = K(\mathbb{Z}_2, 1)$. This sequence is a fibration, and we have the following:

Theorem B.8. *The cohomology ring $H^*(BSO_n; \mathbb{Z}_2)$ is a $\mathbb{Z}_2$ -polynomial ring*

$$\mathbb{Z}_2[w_2, \dots, w_n]$$

where w_k denotes the lift of the universal k th Stiefel-Whitney class by the map $BSO_n \rightarrow BO_n$.

Finally we consider the space $BSpin_n$ which sits in a fibration $BSpin_n \rightarrow BSO_n \rightarrow K(\mathbb{Z}_2, 2)$.

REMARK B.9. The cohomology ring $H^*(BSpin_n; \mathbb{Z}_2)$ is quite complex and since its precise structure is not used here, we refer the interested reader to the basic paper of D. Quillen [1]. (The "stable" groups $H^*(BSpin; \mathbb{Z}_2)$ were first computed by E. Thomas [1].)

The $\mathbb{Z}_2$ -cohomology of the spaces BU_n and BSp_n is merely the mod 2 reduction of the integral cohomology, and we have the following:

Theorem B.10. *The cohomology ring $H^*(BU_n; \mathbb{Z})$ is a $\mathbb{Z}$ -polynomial ring*

$$\mathbb{Z}[c_1, c_2, \dots, c_n]$$

on canonical generators $c_k \in H^{2k}(BU_n; \mathbb{Z})$ for $k = 1, \dots, n$.

The class c_k is called the universal **k th Chern class**. Thus to any n -dimensional complex vector bundle $E \rightarrow X$ classified by a map $f_E: X \rightarrow BU_n$, we have the associated k th Chern class $c_k(E) = f_E^*(c_k)$. There is the concept of the **total Chern class** $c = 1 + c_1 + c_2 + \dots + c_n$, and again there is a product formula

$$c(E \oplus E') = c(E) \cup c(E'). \quad (\text{B.9})$$

Given any complex manifold X , the tangent bundle is a complex vector bundle. We call $c_k(X) \equiv c_k(TX)$ the **k th Chern class** of X .

There is now a ring isomorphism $H^*(BU_n; \mathbb{Z}_2) \cong \mathbb{Z}_2[\tilde{c}_1, \dots, \tilde{c}_n]$ where $\tilde{c}_k$ is of degree $2k$ for each k . The class $\tilde{c}_k$ is the mod 2 reduction of c_k . The natural inclusion $U_n \subset O_{2n}$ induces a map $BU_n \rightarrow BO_{2n}$. It can be shown that under this map $w_{2k} \mapsto \tilde{c}_k$ and, of course, $w_{2k+1} \rightarrow 0$. Hence, for any complex vector bundle E ,

$$w_{2k}(E) \equiv c_k(E) \pmod{2} \quad \text{and} \quad w_{2k+1}(E) = 0$$

for all k .

The picture is much the same for quaternion bundles.

Theorem B.11. *The cohomology ring $H^*(BSp_n; \mathbb{Z})$ is a $\mathbb{Z}$ -polynomial ring*

$$\mathbb{Z}[\sigma_1, \dots, \sigma_n]$$

on canonical generators $\sigma_k \in H^{4k}(BSp_n; \mathbb{Z})$ for $k = 1, \dots, n$.

Under the natural map $BSp_n \rightarrow BU_{2n}$ we have that $c_{2k} \mapsto \sigma_k$ and $c_{2k+1} \mapsto 0$.

The integral cohomology of BO_n is a more complicated story due to the presence of 2-torsion. However, away from the prime 2, things become nice.

Theorem B.12. *Let Λ be any integral domain containing $\frac{1}{2}$ (e.g., $\mathbb{Z}[\frac{1}{2}]$ or $\mathbb{Q}$). Then $H^*(BO_n; \Lambda)$ is the polynomial algebra*

$$\Lambda[p_1, \dots, p_{[n/2]}]$$

on canonical generators $p_k \in H^{4k}(BO_n; \Lambda)$ for $k = 1, \dots, [n/2]$.

Choose $\Lambda = \mathbb{Q}$. Then the class p_k is called the universal k th **rational Pontrjagin class**. Given a real vector bundle $E \rightarrow X$ classified by $f_E: X \rightarrow BO_n$, we define the k th rational Pontrjagin class of E to be $p_k(E) = f_E^*(p_k)$. There is a **total rational Pontrjagin class** $p = 1 + p_1 + \dots + p_{[n/2]}$ and a product formula

$$p(E \oplus E') = p(E) \cup p(E') \quad (\text{B.10})$$

in $H^*(X; \mathbb{Q})$. There are rational Pontrjagin classes of a smooth manifold which, in the compact case, are homeomorphism invariants.

To understand the integral case we must examine the Bockstein homomorphism. The coefficient sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}_2 \rightarrow 0$ gives rise to a long exact sequence

$$\dots \rightarrow H^k(X; \mathbb{Z}) \xrightarrow{2} H^k(X; \mathbb{Z}) \rightarrow H^k(X; \mathbb{Z}_2) \xrightarrow{\beta} H^{k+1}(X; \mathbb{Z}) \rightarrow \dots$$

where β is called the **Bockstein**. Note that the kernel of β is the set of classes in $H^*(X; \mathbb{Z}_2)$ which are mod 2 reductions of integral classes.

Theorem B.13. *The integral cohomology $H^*(\mathrm{BO}_n; \mathbb{Z})$ is an additive direct sum*

$$\mathbb{Z}[p_1, \dots, p_{[n/2]}] \oplus \text{Image}(\beta)$$

where $p_k \in H^{4k}(\mathrm{BO}_n; \mathbb{Z})$ becomes the element p_k of Theorem B.12 under tensor product with Λ , and where β is the Bockstein homomorphism.

The class p_k is called the **universal k th integral Pontrjagin class**. These classes have the following property. Consider the homomorphism $O_n \rightarrow U_n$ induced by complexification. The Chern classes lift back over the induced map $\mathrm{BO}_n \rightarrow \mathrm{BU}_n$. The classes c_{2k+1} go to zero, and

$$p_k = (-1)^k c_{2k}. \quad (\text{B.11})$$

It is conventional to define the Pontrjagin classes of a real vector bundle E by this formula, i.e., by setting

$$p_k(E) = (-1)^k c_{2k}(E \otimes \mathbb{C}). \quad (\text{B.12})$$

Observe now that the subset $\beta(H^*(\mathrm{BO}_n; \mathbb{Z}_2)) \subset H^*(\mathrm{BO}_n; \mathbb{Z})$ consists entirely of 2-torsion elements. Important among these are the classes

$$W_k \equiv \beta(w_{k-1}) \quad (\text{B.13})$$

which measure whether the $(k-1)$ st Stiefel-Whitney class is the mod 2 reduction of an integral class. Of course all the W_k 's vanish when pulled back to BU_n or BSp_n under the maps $\mathrm{BU}_n \rightarrow \mathrm{BO}_{2n}$ and $\mathrm{BSp}_n \rightarrow \mathrm{BO}_{4n}$.

We now examine BSO_n .

Theorem B.14. *Let Λ be an integral domain containing $\frac{1}{2}$. Then for each n , there are ring homomorphisms*

$$\begin{aligned} H^*(\mathrm{BSO}_{2n+1}; \Lambda) &= \Lambda[p_1, \dots, p_n] \\ H^*(\mathrm{BSO}_{2n}; \Lambda) &= \Lambda[p_1, \dots, p_n, \chi] / \langle \chi^2 - p_n \rangle \end{aligned}$$

where $p_k \in H^{4k}(\mathrm{BSO}_n; \Lambda)$ and where $\chi \in H^n(\mathrm{BSO}_{2n}; \Lambda)$.

The elements p_k are the images of the Λ -Pontrjagin classes (of Theorem B.12) under the map $\mathrm{BSO}_n \rightarrow \mathrm{BO}_n$. The class χ is called the **universal Euler class**. It can be defined as usual for any oriented vector bundle $E \rightarrow X$ by setting $\chi(E) = f_E^*(\chi)$. (For odd-dimensional bundles it is defined to be zero.) There is a product formula

$$\chi(E \oplus E') = \chi(E) \cup \chi(E'). \quad (\text{B.14})$$

There is a natural definition of χ as an integral cohomology class. (see Milnor-Stasheff [1]). Under the map $\mathrm{BU}_n \rightarrow \mathrm{BSO}_{2n}$, the class χ pulls back to c_n . Under mod 2 reduction, χ becomes w_n in $H^n(\mathrm{BSO}_{2n}; \mathbb{Z}_2)$.

The analogue of Theorem B.13 holds for $H^*(\mathrm{BSO}_n; \mathbb{Z})$.